

Question 1: Library Book Management System with Exception Handling

Problem Statement:

Design a **Library Book Management System** in Java that:

1. Creates a Book class with attributes:
 - a. bookTitle, bookId, authorName, price, availableCopies
2. Implements custom exceptions for:
 - a. Invalid bookId (not 6 digits)
 - b. Invalid price (≤ 0)
 - c. Book not available ($\text{availableCopies} \leq 0$)
3. Validates book details in a method validateBook().
 - a. Use throws to declare exceptions that must be handled by the caller.
4. Adds validated books to an ArrayList<Book>.
5. Displays all book details **before and after sorting**.
6. Sorts books by:
 - a. **Book Title** descending
 - b. **Price** ascending if titles match
7. Handles all exceptions in the main method.

Question 2: Method Using throws Keyword

Problem Statement:

Write a Java program with a method calculateDivision(int a, int b) that:

1. Performs integer division a / b .
2. Throws an ArithmeticException if b is 0.
3. Declares the exception using throws in the method signature.
4. Calls this method from main() and handles the exception using try-catch.

1)

```
1  import java.util.*;|
2  public class Main {
3      static class Book { 9 usages
4          String title; 7 usages
5          String id; 2 usages
6          String b_name; 1 usage
7          String author_name; 1 usage
8          int price; 6 usages
9          int n_copies; 2 usages
10
11         Book(String t, String i, String b, String a, int p, int n) { 1 usage
12             title = t;
13             id = i;
14             b_name = b;
15             author_name = a;
16             price = p;
17             n_copies = n;
18         }}
19     static class InvalidBookId extends Exception { 3 usages
20         InvalidBookId(String message) { 1 usage
21             super(message);
22         }}
23     static class InvalidPrice extends Exception { 3 usages
24         InvalidPrice(String message) { 1 usage
```

```
2 public class Main {
3     static class InvalidPrice extends Exception { 3 usages
4         InvalidPrice(String message) { 1 usage
5             super(message);
6         }}
7
8     static class BookNotAv extends Exception { 3 usages
9         BookNotAv(String message) { 1 usage
10             super(message);
11         }}
12
13     @ static void validate(Book b) throws InvalidBookId, InvalidPrice, BookNotAv { 1 usage
14         if (b.id.length() != 6)
15             throw new InvalidBookId("Invalid Id");
16         if (b.price <= 0)
17             throw new InvalidPrice("Invalid Price");
18         if (b.n_copies <= 0)
19             throw new BookNotAv("Book Not Available");
20     }
21
22     public static void main(String args[]) {
23         Scanner sc = new Scanner(System.in);
24         ArrayList<Book> list = new ArrayList<>();
25         System.out.print("How many books: \n");
26         int n = sc.nextInt();
27         sc.nextLine();
28
29         ArrayList<Book> list = new ArrayList<>();
30         System.out.print("How many books: \n");
31         int n = sc.nextInt();
32         sc.nextLine();
33         for (int i = 0; i < n; i++) {
34             try {
35                 System.out.println("Enter details for book " + (i + 1));
36                 System.out.print("Title: ");
37                 String title = sc.nextLine();
38                 System.out.print("Book id: ");
39                 String id = sc.nextLine();
40                 System.out.print("Book name: ");
41                 String b_name = sc.nextLine();
42                 System.out.print("Author name: ");
43                 String author_name = sc.nextLine();
44                 System.out.print("Price: ");
45                 int price = sc.nextInt();
46                 System.out.print("Copies: ");
47                 int n_copies = sc.nextInt();
48                 sc.nextLine();
49                 Book b = new Book(title, id, b_name, author_name, price, n_copies);
50                 validate(b);
51             }
52         }
53     }
54 }
```

```
62         Book b = new Book(title, id, b_name, author_name, price, n_copies);
63         validate(b);
64         list.add(b);
65         System.out.println("Book added");
66     } catch (InvalidBookId | InvalidPrice | BookNotAv e) {
67         System.out.println("Error: " + e.getMessage());
68         i--; // again try book
69         sc.nextLine();
70     }}
71     System.out.println("Before sorting: \n");
72     for (int i = 0; i < list.size(); i++) {
73         Book b = list.get(i);
74         System.out.println(b.title + " " + b.price);
75     }
76     Collections.sort(list, new Comparator<Book>() {
77     i↑
78         public int compare(Book b1, Book b2) {
79             if (!b1.title.equals(b2.title))
80                 return b2.title.compareTo(b1.title); // descending title
81
82             System.out.println(b.title + " " + b.price);
83         }
84     });
85     Collections.sort(list, new Comparator<Book>() {
86     i↑
87         public int compare(Book b1, Book b2) {
88             if (!b1.title.equals(b2.title))
89                 return b2.title.compareTo(b1.title); // descending title
90             return b1.price - b2.price; // ascending price
91         }
92     });
93     System.out.println("\nAfter sorting:");
94     for (Book b : list) {
95         System.out.println(b.title + " " + b.price);
96     }
97     sc.close();
98 }}
99
```

Output 1:

Submitted By: Arpita Rani

```
How many books:
3
Enter details for book 1
Title: a
Book id: 789087
Book name: t
Author name: u
Price: 90
Copies: 9
Book added
Enter details for book 2
Title: u
Book id: 89
Book name: y
Author name: i
Price: 45
Copies: -2
Error: Invalid Id
```

Output 2:

Submitted To: Nipun Gera

Submitted By: Arpita Rani

```
How many books:
3
Enter details for book 1
Title: a
Book id: 834682
Book name: q
Author name: w
Price: 45
Copies: 4
Book added
Enter details for book 2
Title: w
Book id: 342466
Book name: r
Author name: t
Price: 89
Copies: 9
Book added
Enter details for book 3
Title: y
Book id: 983468
```

Submitted To: Nipun Gera

Submitted By: Arpita Rani

```
Book added
Enter details for book 3
Title: y
Book id: 983468
Book name: u
Author name: i
Price: 78
Copies: 7
Book added
Before sorting:

a 45
w 89
y 78

After sorting:
y 78
w 89
a 45

Process finished with exit code 0
```

Submitted To: Nipun Gera

Submitted By: Arpita Rani

Ans 2:

```
3  ▶ public class Main {
4      public static int division(int a , int b) throws ArithmeticException 2 usages
5      {
6          if(b==0)
7          { throw new ArithmeticException("denominator can't be zero");
8          }
9          return a/b;
10     }
11  ▶ public static void main(String[] args) {
12      try{
13          int r=division( a: 18, b: 9);
14          System.out.println("Answer is : "+r);
15      int r2=division( a: 18, b: 0);
16          System.out.println("Answer is : "+r2);
17      }
18      catch(ArithmeticException e)
19      { System.out.println("Error: "+e.getMessage());}
20  }
21 }
```

Microsoft Defender may affect IDE

Output:

```
"C:\Program Files\Java\jdk-11\bin\java.exe
Answer is : 2
Error: denominator can't be zero

Process finished with exit code 0
|
```

Submitted To: Nipun Gera